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Department of Electrical & Electronic Engineering
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Course Number : EEE282

Section :**01**

Group No : **04**

Lab Report

Experiment no: 01

Name of the experiment:

Introduction of MATLAB: Familiarizing of MATLAB

Prepared by:

Name:Souvik Barman Ratul ID: 24121205

All Group members:

Sl.	*ID*	*Name*
1.	24121205	Souvik Barman Ratul
2.	24121308	Alif Tamjid
3.	23221082	Sabrina Maliha
4.	24121019	Monazzi Siddique

(1)

Objective:

In this experiment, we can learn the fundamental uses of MATLAB for performing diverse mathematical operations on data structures like scalars, vectors, matrices, complex numbers, indexing arrays, and applying essential programming constructs to generate and plot functions.

Theoretical Background:

This program shows MATLAB's fundamental structure of numerical data on vectors and matrices. Here,we have to do many mathematical operations like element-wise operations (like .* and .^) and matrix multiplication (*). The use of transpose (.') helps in converting row vectors to column vectors when needed. Familiarity with built-in mathematical functions such as sqrt(), mean(), and size() is important for performing calculations and statistical analysis. Additionally, we use some commands like fprintf or disp for presenting outputs in a clear and formatted way.

Problem Statement:

No. 1:

● Put the digits of your ID’s in a row vector.

● Define another vector which has elements 1 to 8.

● Multiply the two vectors of step 1 & 2 element by element to get the vector element_mult.

● Multiply the vector of step 1 & transpose of the vector of step 2 normally to get the vector norm_mult.

● Now multiply element_mult & norm_mult to find final_value. ● Find the square root of the final value and define this vector as root_fv

● Obtain the size of the vector element_mult to get the vector sz

● Use the last element of the id vector as the power of each element of root_fv. Thus find the vector p. Those having 0 as the last digit of id should use the 2nd last digit instead.

● Define a vector sum, which is the summation of vector p and the 2nd element of the size vectors z.

● Find the mean of the vector sum. Display all the values.

Answer:

<u>Code:</u>

```
clear all;

close all;

clc;

ID = [2 4 1 2 1 2 0 5];

A = [1:8];


element_mult = ID.*A ;

H = A.' ;   % transpose of step 2

norm_mult = ID*H;

final_value = element_mult.*norm_mult;

root_fv = sqrt(final_value);

sz= size(element_mult);

P = root_fv.^ 5;  % Last element of ID is 5

T = sz(2);    % 2nd size element

vector_sum = P + T;
```

```
Mean = mean(vector_sum);

fprintf('\nID = '); fprintf('%g ', ID); fprintf('\n');

fprintf('\nA = '); fprintf('%g ', A); fprintf('\n');

fprintf('\nElement-wise Multiplication (element_mult) = '); fprintf('%g ', element_mult); fprintf('\n');

fprintf('\nTranspose of A (H) = '); fprintf('%g ', H); fprintf('\n');

fprintf('\nNormal Multiplication (norm_mult) = %g\n', norm_mult);

fprintf('\nFinal Value (final_value) = '); fprintf('%g ', final_value); fprintf('\n');

fprintf('\nSquare Root of Final Value (root_fv) = '); fprintf('%.4f ', root_fv); fprintf('\n');

fprintf('\nSize of element_mult (sz) = '); fprintf('%g ', sz); fprintf('\n');

fprintf('\nPower Value (P) = '); fprintf('%.4e ', P); fprintf('\n');

fprintf('\nVector Sum = '); fprintf('%.4e ', vector_sum); fprintf('\n');

fprintf('\nMean of Vector Sum = %.4f\n', Mean);
```

Screenshot of the code:

Code_1.m | Code_2.m | code_3.m

```
clear all;
close all;
clc;
ID = [2 4 1 2 1 2 0 5];
A = [1:8];

element_mult = ID.*A ;

H = A.' ;   % transpose of step 2

norm_mult = ID*H;

final_value = element_mult.*norm_mult;

root_fv = sqrt(final_value);

sz= size(element_mult);

P = root_fv.^ 5;  % Last element of ID is 5

T = sz(2);    % 2nd size element

vector_sum = P + T;

Mean = mean(vector_sum);

fprintf('\nID = '); fprintf('%g ', ID); fprintf('\n');
fprintf('\nA = '); fprintf('%g ', A); fprintf('\n');
fprintf('\nElement-wise Multiplication (element_mult) = '); fprintf('%g ', element_mult); fprintf('\n');
fprintf('\nTranspose of A (H) = '); fprintf('%g ', H); fprintf('\n');
fprintf('\nNormal Multiplication (norm_mult) = %g\n', norm_mult);
fprintf('\nFinal Value (final_value) = '); fprintf('%g ', final_value); fprintf('\n');
fprintf('\nSquare Root of Final Value (root_fv) = '); fprintf('%.4f ', root_fv); fprintf('\n');
fprintf('\nSize of element_mult (sz) = '); fprintf('%g ', sz); fprintf('\n');
fprintf('\nPower Value (P) = '); fprintf('%.4e ', P); fprintf('\n');
fprintf('\nVector Sum = '); fprintf('%.4e ', vector_sum); fprintf('\n');
fprintf('\nMean of Vector Sum = %.4f\n', Mean);
```

Figure 1: Main code of element wise ,transpose,matrix and other operations.

Output:

```
Command Window

ID = 2 4 1 2 1 2 0 5

A = 1 2 3 4 5 6 7 8

Element-wise Multiplication (element_mult) = 2 8 3 8 5 12 0 40

Transpose of A (H) = 1 2 3 4 5 6 7 8

Normal Multiplication (norm_mult) = 78

Final Value (final_value) = 156 624 234 624 390 936 0 3120

Square Root of Final Value (root_fv) = 12.4900 24.9800 15.2971 24.9800 19.7484 30.5941 0.0000 55.8570

Size of element_mult (sz) = 1 8

Power Value (P) = 3.0396e+05 9.7266e+06 8.3761e+05 9.7266e+06 3.0037e+06 2.6803e+07 0.0000e+00 5.4373e+08

Vector Sum = 3.0396e+05 9.7266e+06 8.3761e+05 9.7266e+06 3.0037e+06 2.6803e+07 8.0000e+00 5.4373e+08

Mean of Vector Sum = 74266994.5072
fx >>
```

Figure 1(i): Output of the given directions.

Explanation:

In this code firstly input my ID number and put its digits into a vector. Then it creates another vector A with numbers 1 to 8. Both vectors are multiplied element by element to get element_mult. Next, the code multiplies the ID vector with the transpose of A to get norm_mult. Then both results are multiplied to get final_value, and its square root is found as root_fv. The last digit of your ID (5) is used as a power to raise each element of root_fv to the 5th power, giving P. After that, we know the size of element_mult is [1 8]. So the 2nd element 8 is added to each element to make vector_sum. Finally, the code finds the mean value of that vector and displays all the results using fprintf.

Discussion:

The experiment developed a MATLAB program to perform a series of vector and matrix operations using built-in arithmetic and mathematical functions. The program initialized two arrays, ID and A, and carried out element-wise and normal multiplications to demonstrate the difference between these operations in MATLAB. It then applied functions such as transpose, square root, power, and mean to process and analyze the resulting vectors. The final output displayed intermediate and final results in the command window using formatted print statements. Through this experiment, the concept of vectorized computation, matrix transposition, and the distinction between element-wise (.*) and matrix (*) operations were strong. The exercise also provided practical experience with MATLAB's array manipulation and output formatting. Overall, the experiment enhanced understanding of basic array operations and mathematical functions in MATLAB, building a foundation for more advanced numerical and matrix-based programming tasks.

(2)

Objective:

The aim of this experiment is to generate sine wave and cosine waves of different amplitudes and frequencies and understand the use of pot and subplot commands.

Theoretical Background:

Sine waves and cosine waves are two fundamental periodic functions. It is used to represent the oscillatory signals in maths and engineering. They have the general formula Asin(2pi ft) and Acos(2pi ft) where A stands for Amplitude, f stands for frequency(Hz) and t stands for time.

In this experiment, MATLAB is used to make sine waves and cosine wave with different frequencies and amplitudes, and learn the use of plot commands (to make graphs) and subplot commands (allows multiple plots on the same window). This experiment allows us to generate signals with different frequencies and visualize it with the help of graphs.

Problem Statement:

- Define a vector t which has values from 0 to 80 ms with an interval of 0.001
- Make two sine and one cosine waves with frequencies 15 & 30 & 60 and amplitudes 10, 2 & 5 respectively.
- Plot the two sine waves in two different figure windows.
- Again plot the two sine waves in the same figure window.
- Using the subplot command, plot all of them on the same window, but different plots.
- Using the subplot command, plot all the sine waves on the same figure and cosine in a different plot.

Answer:

<u>Code:</u>

```
clear all;
close all;
clc;
t = 0./1000:0.001:80./1000;       %vector t with values from 0 to 80ms and 0.001
interval of time.
S1 = 10.* sin(2.*pi.*15.*t);          %Defining sine wave with frequency 15 and
amplitude 10.
S2 = 2.* sin(2.*pi.*30.*t);             %Defining sin wave with frequency 30 and
amplitude 2.
C1 = 5.* cos(2.*pi.*60.*t);             %Defining cos wave with frequency 60 and
amplitude 5.

figure (1)                               %it is used to get figures in different windows.
plot (t,S1,'b');       %plot is used to make a sine wave(S1) vs time graph,b indicates blue color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                       %to get grid lines on the graph.

figure (2)                    %it is used to get figures in different windows.
plot (t,S2,'r');         %plot is used to make a sine wave(S2) vs time graph,r indicates
red color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                   %to get grid lines on the graph.

figure(3)              %it is used to get figures in different windows.
plot(t,S1,'b');      %plot is used to make a sine wave(S1) vs time graph,b indicates
blue color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                %to get grid lines on the graph.
hold on;                % to save the previous graph result.
```

```
plot (t,S2,'r');        %plot is used to make a sine wave(S2) vs time graph,r indicates
red color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;%to get grid lines on the graph.
hold on;% to save the previous graph result.

figure (4)              %it is used to get figures in different windows.
subplot(2,2,1);         % same window,to get the graph in first column first row.
plot (t,S1,'b');        %plot is used to make a sine wave(S1) vs time graph,b indicates
blue color.
xlabel('Times(ms)');
ylabel('Amplitude');
title('Sin wave(S1) with f=15 & A=10');   %used to write the title of graph.
grid on;                                  %to get grid lines on the graph.
hold on;                                  % to save the previous graph result.

subplot(2,2,2);  % same window,to get the graph in second column first row.
plot (t,S2,'r');   %plot is used to make a sine wave(S2) vs time graph,r indicates red
color.
xlabel('Times(ms)');
ylabel('Amplitude');
title('Sin wave(S2)with f=30 & A=2');
grid on;          %to get grid lines on the graph.
hold on;          % to save the previous graph result.


figure(5)               %it is used to get figures in different windows.
subplot(2,1,1);         % same window,to get the graph in first column first row.
plot(t,S1,'b');       %plot is used to make a sine wave(S1) vs time graph,b indicates blue color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                %to get grid lines on the graph.
hold on;                % to save the previous graph result.
plot(t,S2,'r');         %plot is used to make a sine wave(S2) vs time graph,r indicates red color.
```

```
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                %to get grid lines on the graph.
hold on;                % to save the previous graph result.
title('Sin wave with f=15,30 & A=10,2');    %used to write the title of graph.
subplot(2,1,2);         % same window,to get the graph in first column first row.
plot(t,C1);             %plot is used to make cos wave(C1) vs Time(ms).
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                      %to get grid lines on the graph.
title('Cos wave with f=60 & A=5');      %used to write the title of graph.
```

Screenshot of input code (from MATLAB online version):

Exp2.m × +

/MATLAB Drive/Exp2.m

```
clear all;
close all;
clc;
t = 0./1000:0.001:80./1000;        %vector t with values from 0 to 80ms and 0.001 interval of time.
S1 = 10.* sin(2.*pi.*15.*t);         %Defining sine wave with frequency 15 and amplitude 10.
S2 = 2.* sin(2.*pi.*30.*t);           %Defining sin wave with frequency 30 and amplitude 2.
C1 = 5.* cos(2.*pi.*60.*t);          %Defining cos wave with frequency 60 and amplitude 5.

```

Figure 2(i):

```

figure (1)                                       %it is used to get figures in different windows.
plot (t,S1,'b');        %plot is used to make a sine wave(S1) vs time graph,b indicates blue color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                           %to get grid lines on the graph.

figure (2)                       %it is used to get figures in different windows.
plot (t,S2,'r');        %plot is used to make a sine wave(S2) vs time graph,r indicates red color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                 %to get grid lines on the graph.
```

Figure 2(ii):

```
figure(3)              %it is used to get figures in different windows.
plot(t,S1,'b');      %plot is used to make a sine wave(S1) vs time graph,b indicates blue color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                %to get grid lines on the graph.
hold on;               % to save the previous graph result.
plot (t,S2,'r');        %plot is used to make a sine wave(S2) vs time graph,r indicates red color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;%to get grid lines on the graph.
hold on;% to save the previous graph result.
```

Figure 2(iii)

Exp2.m

/MATLAB Drive/Exp2.m

```
figure (4)              %it is used to get figures in different windows.
subplot(2,2,1);        % same window,to get the graph in first column first row.
plot (t,S1,'b');        %plot is used to make a sine wave(S1) vs time graph,b indicates blue color.
xlabel('Times(ms)');
ylabel('Amplitude');
title('Sin wave(S1) with f=15 & A=10');   %used to write the title of graph.
grid on;                                              %to get grid lines on the graph.
hold on;                                             % to save the previous graph result.

subplot(2,2,2);  % same window,to get the graph in second column first row.
plot (t,S2,'r');   %plot is used to make a sine wave(S2) vs time graph,r indicates red color.
xlabel('Times(ms)');
ylabel('Amplitude');
title('Sin wave(S2)with f=30 & A=2');
grid on;           %to get grid lines on the graph.
hold on;            % to save the previous graph result.
```

Figure 2(iv)

```
figure(5)               %it is used to get figures in different windows.
subplot(2,1,1);         % same window,to get the graph in first column first row.
plot(t,S1,'b');       %plot is used to make a sine wave(S1) vs time graph,b indicates blue color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                    %to get grid lines on the graph.
hold on;                   % to save the previous graph result.
plot(t,S2,'r');         %plot is used to make a sine wave(S2) vs time graph,r indicates red color.
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;               %to get grid lines on the graph.
hold on;              % to save the previous graph result.
title('Sin wave with f=15,30 & A=10,2');    %used to write the title of graph.
subplot(2,1,2);         % same window,to get the graph in first column first row.
plot(t,C1);           %plot is used to make cos wave(C1) vs Time(ms).
xlabel('Times(ms)');
ylabel('Amplitude');
grid on;                         %to get grid lines on the graph.
title('Cos wave with f=60 & A=5');      %used to write the title of graph.
```

Figure 2(v)

Output:-

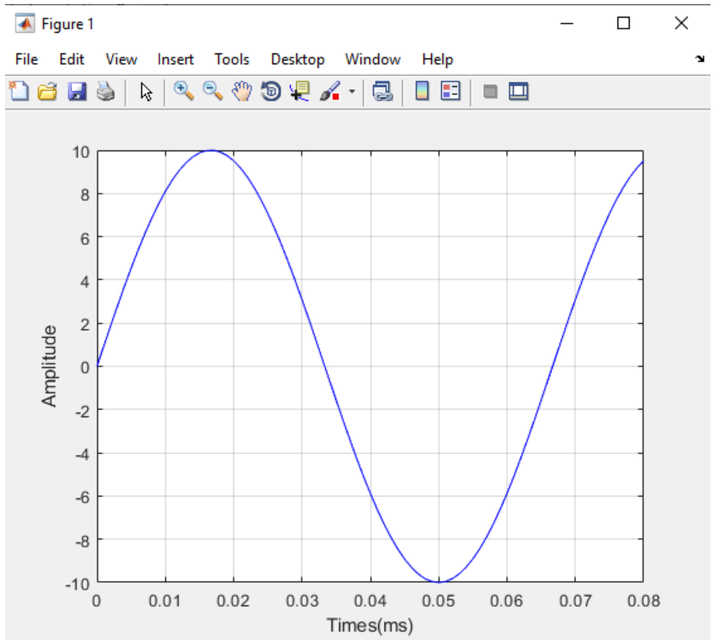

Figure 2(vi): Sine wave frequency 15 and Amplitude 10.

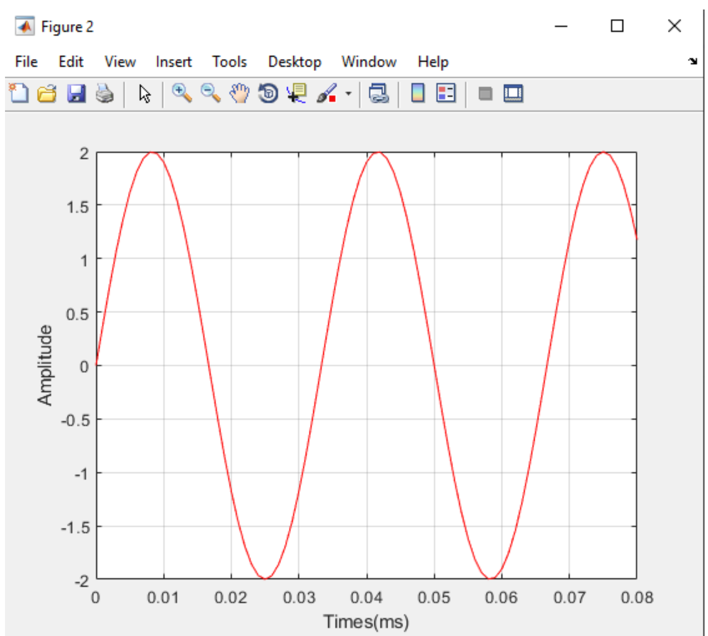

Figure 2(vii): Sine wave frequency 30 and Amplitude 2.

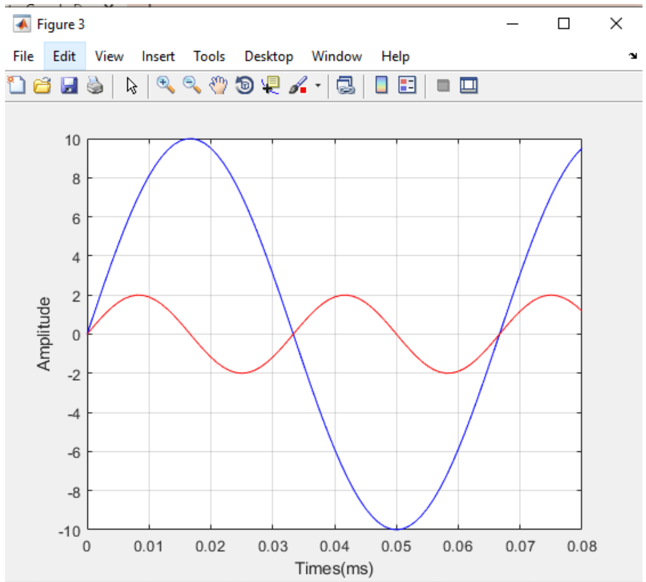

Figure 2(viii): Two sine waves in the same plots using “hold on” function.

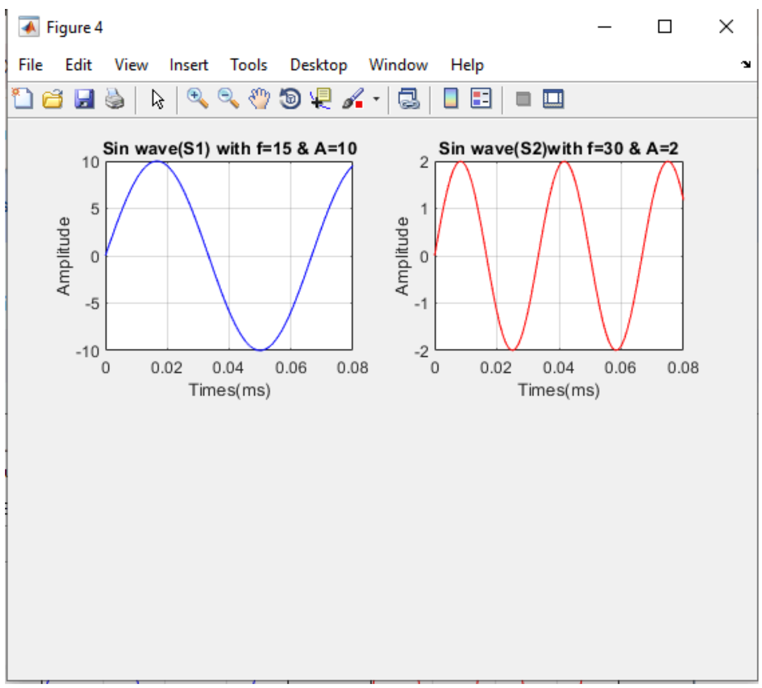

Figure 2(ix): Two sine waves in the same window but different plots using “subplot” function.

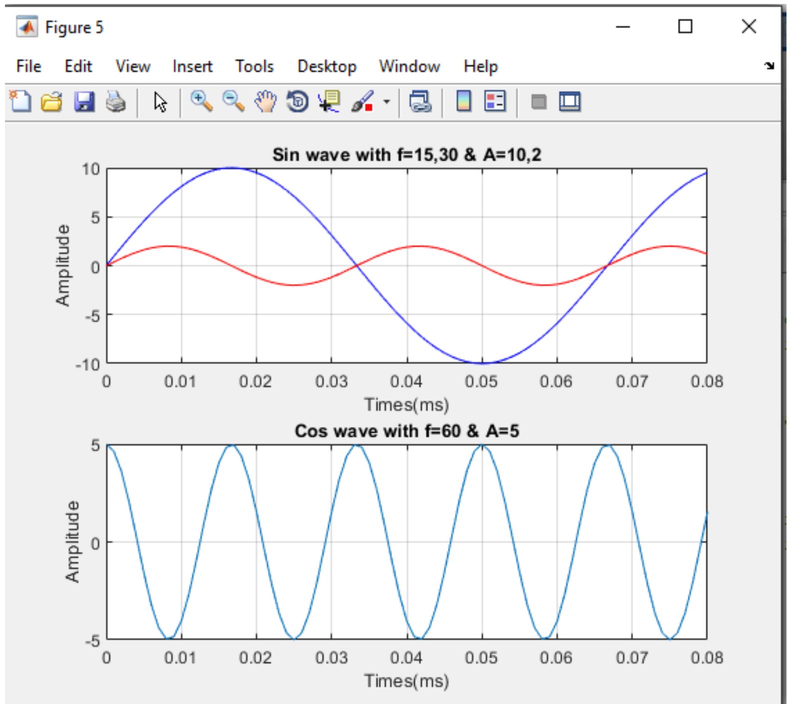

Figure 2(x): Two sine waves are in the same plots because of using the “hold on” function and by using the “subplot” function, cosine wave is plotted in the same window but in a different plot.

Explanation:

In the main script of the code, various items in workspace and memory are cleared by using clear all, close all and clc. A time vector t from 0 to 80 ms was taken with an interval of 0.001. Then two sine wave functions with frequency(f) 15 and 30 and amplitude(A) 10 and 2 were taken respectively. And a cosine wave with frequency 60 and amplitude 5 was taken. Following the general formula of sine wave [Asin(2pi ft)] and cosine wave [Acos(2pi ft)], the waves were defined.

“figure ( )” function was used to get the graph of two waves in two separate windows. “plot” function was used to get the graph that has time in x axis and amplitude in y axis. “xlabel ( )” and “ylabel ( )” were used to label the x and y axis as amplitude and time respectively. “grid on” function was used to get grid lines on the graph.

In figure 3, “hold on” function was used to keep the previous graph of the first sine wave intact and to draw both the sine waves on the same graph.

In figure 4, “subplot ( )” command was used to draw all the plots in the same window but in different columns.The “subplot” command used here had 2 rows and 2 columns. And the “title ( )” command is used here to give the title over the plot.

In figure 5 , “subplot ( )” command was used to draw two sine waves and a cosine wave in the same window. Two sine waves were drawn on the same plot. That's why the “hold on” command was also used here. But no “hold on" command is used in case of cos wave. It is because we want to plot it separately on the same window. Moreover, many other commands were added to give color and others to enhance the beauty of the graph.

Discussion:

In this experiment, sine wave and cosine waves with different amplitudes and frequencies were generated and plotted using MATLAB. A time vector was used whose initial values were mentioned as well as its interval of 0.001 to get a smooth, continuous graph. The 2 sine waves and a cosine wave were created using MATLAB’s built in function.

The signals were first potted in separate windows and then in the same plot, two sine waves were drawn. Then two waves are drawn in the same window but with different plots. According to the last directive, the two waves were plotted in the same graph in the same window but the cosine wave was drawn on a different plot in the same window.